
Supplementary Material for: Causally-Faithful Radiology Report Generation
via Dual-Adversarial Counterfactual Training

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

A. Reproducibility Checklist

- [✓] Training data sources are public (OpenI for training and in-distribution evaluation; ChestX-Det10 silver-target for cross-site eval; PadChest-GR named as deferred cross-site target; MIMIC-CXR is not used in this study).
- [✓] Hyperparameters specified per-config in YAML; full hyperparameter table in §B.4.
- [✓] 30 unit tests pass on CPU; full test list in §B.5.
- [✓] Modal infrastructure described; all entrypoints documented.
- [✓] Per-fine-tune training-step counts, save-every-steps schedule, and resume-from-checkpoint logic released as code.
- [✓] Cross-site IMG_correct on ChestX-Det10 silver-target measured (§??).
- [] Code release URL — anonymized for double-blind review; will be filled at camera-ready.
- [] Trained model weights release URL — anonymized for double-blind review; subject to MedGemma research-only redistribution terms.
- [] Cross-site IMG_correct on PadChest-GR (gold or silver) — see §??; deferred to follow-up.
- [] Three-seed averaging on noise/crop/occlude cross-counterfactuals — single-seed in §??; deferred to follow-up.
- [] Mech-interp sweep validating the gradient-norm monitor’s AUROC — deferred to follow-up.

B. Extended Methods

B.1. Faith loss derivation, edge cases, and implementation

The naive consistency loss is $\mathcal{L}_{\text{KL}} = \text{ReLU}(\text{margin} - \text{KL}(p_{\text{full}} \| p_{\text{alt}}))$. This admits a degenerate optimum: a model that always emits a uniform distribution under masking satisfies any finite KL margin without using the image. The token-wise correctness-gain margin replaces the distributional divergence with a per-token signal: $\mathcal{L}_{\text{faith}}(x) = (1/|T|) \sum_{t \in T} \text{ReLU}(\text{margin} - (\log p_{\text{full}}(y_t) - \log p_{\text{alt}}(y_t)))$, which couples the faithfulness requirement to gold-token correctness on the full input. The model cannot satisfy the margin by being arbitrarily wrong on the image-zeroed input.

The implementation runs three forward passes per training step — full, image-zeroed, and image-flipped — and exposes a `flipped_row_mask` that zeroes the flipped-counterfactual contribution on rows that are not laterality-sensitive (claims that do not contain one of left, right, bilateral). Image-zeroing is implemented as the per-channel mean of the input batch’s pixel values, not a literal zero tensor: a literal zero is, after the processor’s normalisation, a specific out-of-distribution input the model has never seen at training time; the per-channel mean produces a uniform “neutral” image at the model’s own normalisation, which is what we want when we say “the image is absent.” Edge cases: when no rows in the batch are laterality-sensitive, the flipped term contributes zero; when the assistant turn has zero target tokens, the loss is clamped to zero without producing NaN gradients.

B.2. Dual filter four-cell decomposition

After the subsequent text-target filter restricts training to OpenI rows with non-empty findings text, the per-row weight distribution within the surviving 10,839 rows is 95.6% at $w_{\min} = 0.2$ and 4.4% at $w = 1.0$. The continuous-SFC ablation replaces the discrete $\{w_{\min}, 1\}$ rule with $\text{SFC}(x) = 1 - p_{\text{text}}^{\text{correct}}(x)$ clipped to $[w_{\min}, 1]$, which provides a smoother weighting and matches the published SFC recipe.

Table 1. Four-cell decomposition of which filter branch flags each training row. On the 20,560-row training pool, the image-only branch catches an additional 6.5% of rows that the text-only branch alone would miss; this is the symmetric-extension contribution axis relative to ?.

Filter outcome	Count	% of pool
Both filters caught (unimodally solvable on either axis)	15,375	74.8%
Text-only caught, image-only passed	2,156	10.5%
Image-only caught, text-only passed (novelty axis)	1,329	6.5%
Neither caught (genuinely multimodal)	1,700	8.3%

B.3. IMG_correct full derivation, edge cases, and the “IMG_correct > IMG” phenomenon

For a binary classifier f with gold class y on input x , define four mutually-exclusive transitions under image-masking:

- $cc(x)$: correct on full, correct on masked.
- $ci(x)$: correct on full, incorrect on masked (the IMG sub-population).
- $ic(x)$: incorrect on full, correct on masked.
- $ii(x)$: incorrect on both.

Then $IMG = (n_{ci} - n_{ic})/n \times 100$, because IMG measures $acc_{full} - acc_{masked} = ((n_{cc} + n_{ci}) - (n_{cc} + n_{ic}))/n = (n_{ci} - n_{ic})/n$. $IMG_correct$ further partitions ci into a margin-passing subset and a margin-failing subset:

$$n_{ci, no-margin} = |\{i : ci(x_i)\}| \quad (1)$$

$$n_{ci, with-margin} = |\{i : ci(x_i) \wedge p_{full}(y_i) - p_{masked}(y_i) \geq \tau\}| \quad (2)$$

and defines:

$$IMG_correct = n_{ci, with-margin}/n \times 100 \quad (3)$$

$$\text{induced inv.} = (n_{ci, no-margin} - n_{ci, with-margin})/n \times 100. \quad (4)$$

$IMG_correct$ can exceed IMG . When the transition population ic is non-trivial (model became correct under masking) and the margin-failing- ci population is small, $IMG_correct$ is $n_{ci, with-margin}/n$ which can be larger than $(n_{ci} - n_{ic})/n$. This is mathematically why several configurations in our retroactive audit (Table 3, §C) show $IMG_correct > IMG$. It does not indicate negative induced inversion. For autoregressive generators we apply this same logic per token on the gold target sequence, average over assistant- turn token positions only, and require the causal-LM shift between logits and target IDs (logits at position t predict token $t + 1$).

B.4. Training hyperparameters and compute

The faith-loss configurations are $\sim 3\times$ slower per step than simple-loss configurations because each step runs three forward passes (full, image-zeroed, image-flipped). Total Phase 4 fine-tuning compute is $\approx \$200$ on Modal H100 80GB across the 15 fine-tunes (5 configs $\times$ 3 seeds). Total project envelope (training, primary evals, retroactive audit, cross-counterfactual, cross-site) is $\approx \$400$. All experiments are reproducible on a single H100 80GB instance with the released code.

B.5. Test coverage

The released codebase contains 30 unit tests (CPU, $\sim 1s$ wallclock) covering: (i) faith-loss behaviour — zero-loss when full overwhelms alt; positive-loss when equal; SFT-coupling-required scenario; correctness term; padding ignored; weights additive; differentiability; flipped-row-mask filtering; zero-valid-rows no-NaN (tests/test_faith_loss.py, 9 tests). (ii) $IMG_correct$ behaviour — perfect grounding; pure inversion caught at margin; mixed; no-change; shape mismatch raises; generator-shift contract; v5_3 replication scenario (tests/test_img_correct.py, 7 tests). (iii) Dual-filter behaviour — min, product aggregations; mismatched row-sets raises; unknown aggregation raises; disagreement-rate counting (tests/test_dual_filter.py, 5 tests). (iv)

Table 2. Training hyperparameters across all five configurations. The five configurations differ only in (i) which weights JSONL is loaded (if any), (ii) whether the faith loss is enabled, (iii) whether the flipped counterfactual is enabled, and (iv) the resulting λ_{faith} . All other hyperparameters are identical.

Hyperparameter	Value
Backbone	MedGemma-4B-IT (research-only licence)
LoRA rank r	16
LoRA alpha	32
LoRA targets	q_proj, k_proj, v_proj, o_proj
Trainable parameters	11.9 M (0.28% of 4.31 B base)
Optimizer	AdamW
Learning rate	2×10^{-4}
Weight decay	0.0 on LoRA targets, 0.01 elsewhere
LR schedule	cosine to zero, 50-step linear warm-up
Batch size	2
Gradient accumulation	$\times 8$ (effective batch size 16)
Max optimizer steps	2,500
Mixed precision	bf16
Faith-loss margin (nats)	0.3
Image masking	per-channel mean of batch pixel values
λ_{faith}	0.3 when enabled, 0 otherwise
w_{min} (filter downweight)	0.2
Filter activation threshold	0.7 on $p_{\text{branch}}^{\text{correct}}$
Random seeds	{42, 1337, 9001}
GPU	NVIDIA H100 80 GB (Modal)
Wallclock per fine-tune	1.5–4.0 h (faith-loss configs $\sim 3\times$ slower)

Data-loader behaviour — laterality regex; collate padding; three image variants on laterality batch; flipping disabled; no-laterality rows; labels equal input-ids; verifier-only filter drop; weight-remapping with interleaved no-target rows (tests/test_data_loader.py, 9 tests).

C. Full IMG_correct Retroactive Audit

Distribution of induced-inversion contributions. Across all 18 cells: induced inversion has an absolute range of [0.00, 1.31] pp and a relative range of [0.0%, 13.2%] of the corresponding IMG. The cell with the largest relative induced inversion is v5_0_base at 13.2%, with the smallest absolute IMG (1.28 pp); fractionally large because any flipping at all under masking is fractionally large at this scale. The cell with the largest absolute induced inversion is v6_0_loo_no_padchest at 1.31 pp (3.1% of IMG). No cell exceeds the pre-registered $\geq 30\%$ threshold; no cell is even close.

The abl_no_consist cell is the cleanest disconfirmation of the pre-registered concern. Removing the consistency loss collapses IMG_correct from ~ 70 pp (nominal v5 family) to 3.60 pp — near baseline — while simultaneously reducing induced inversion from ~ 0.7 pp to 0.30 pp. The pre-registered concern was that the consistency loss caused induced inversion. The empirical pattern is: with the consistency loss, IMG_correct is high and induced inversion is small; without it, both are small. The consistency loss contributes the IMG_correct, not the induced inversion.

D. Cross-Counterfactual Detail

Baseline anchor. On the matching crop counterfactual, sft_only seed-42 yields IMG = 0.33 pp, IMG_correct = 2.80 pp at 73.82% accuracy, induced inversion = 0.26 pp ($n = 105,921$ tokens). The cross-counterfactual sft_full – sft_only IMG_correct gap on crop is $4.81 - 2.80 = 2.01$ pp; the gap on the trained mean counterfactual is $74.00 - 2.38 = 71.62$ pp (single-seed comparison; the 3-seed comparison gives 72.58 pp). The order-of-magnitude difference between trained-mean and crop is the operational meaning of the $\S??$ partial-circularity finding.

Table 3. Full IMG_correct retroactive audit, all 18 cells (nine verifier configurations + nine v5 ablations, each on $n = 2,974$ verifier rows). Acc full and Acc masked are accuracies under the full and image- zeroed inputs, respectively.

Config	IMG_corr	IMG	Inv	Inv/IMG	Acc full	Acc mask
Headline configurations.						
v5_0_base	3.83	1.28	0.17	13.2%	92.30	91.02
v5_1_ground	3.40	2.62	0.13	5.1%	91.22	88.60
v5_2_real	64.66	61.84	0.91	1.5%	92.60	30.77
v5_3_contrast	71.05	68.12	0.74	1.1%	93.01	24.88
v5_4_final	68.43	63.55	0.40	0.6%	91.69	28.14
v6_0_3site	74.51	70.21	0.07	0.1%	91.06	20.85
v6_0_loo_no_openi	38.20	34.20	0.00	0.0%	87.42	53.23
v6_0_loo_no_padchest	43.95	41.80	1.31	3.1%	92.30	50.50
v6_0_loo_no_chestx	18.59	14.53	0.20	1.4%	81.27	66.75
v5 ablations.						
abl_scale_25	69.07	65.64	0.71	1.1%	92.70	27.07
abl_scale_50	70.51	67.59	0.81	1.2%	92.80	25.22
abl_scale_100	66.24	62.81	0.61	1.0%	92.80	29.99
abl_hothresh_70	73.84	69.97	0.57	0.8%	92.60	22.63
abl_hothresh_80	77.98	74.78	0.74	1.0%	92.77	17.99
abl_no_contrast	63.28	60.02	0.87	1.4%	92.77	32.75
abl_no_ground	72.53	69.64	0.50	0.7%	92.97	23.34
abl_no_uncert	73.40	70.54	0.98	1.4%	93.17	22.63
abl_no_consist	3.60	2.79	0.30	10.8%	88.80	86.01

Table 4. Cross-counterfactual eval — full numbers including masked accuracy. Each row uses the same `sft_full` seed-42 checkpoint; only the counterfactual transformation differs. The masked accuracy column makes visible that under the trained mean and noise counterfactuals the model drives image-zeroed accuracy to near-zero (1.51% and 13.38% respectively), while under crop and occlude the model retains 71.83% and 75.20% accuracy — close to the unperturbed 75.48%. The faith loss has trained the model to drive $p(y | \text{neutral image}) \rightarrow 0$ on the trained counterfactual; it has not trained the model to do so on counterfactuals it has not seen.

Counterfactual	IMG	IMG_corr	Acc full	Acc mask
mean (TRAINED)	73.97	74.00	75.48	1.51
noise	62.10	62.20	75.48	13.38
crop	3.64	4.81	75.48	71.83
occlude	0.28	1.11	75.48	75.20

E. Cross-Site Detail

Silver target construction. The ChestX-Det10 test rows in the standard test set have empty `evidence_text` fields. We constructed `groundbench_test_chestx_det10_silver.jsonl` (1,122 rows) by injecting CheXagent-2-3B (?) generations as `reference_report`; the silver-target nature is reported explicitly. Reviewers may push on this caveat. PadChest-GR cross-site (?) requires a fresh CheXagent generation pass and is deferred. Token-level analysis. The cross-site 54.65% token-level accuracy is 20 pp lower than in-distribution OpenI accuracy, reflecting (i) genuine cross-site difficulty (different camera vendors, different radiologist phrasing) and (ii) the silver-target nature (the model is being graded against another model’s output, not radiologist gold).

F. Pre-Registered Hypotheses

(i) Full method reduces IMG_correct on OpenI-test by ≥ 5 pp vs SFT-only at matched accuracy. Status: HIT BY AN ORDER OF MAGNITUDE on three seeds $\times$ three faith-loss configs. Realised: +72.58 pp at +1.17 pp accuracy.

(ii) IMG_correct refinement reveals $\geq 30\%$ of v5_3_contrast’s IMG as induced inversion. Status: REFUTED. Realised: 1.1%. Pattern holds across all 18 audited cells (induced-inversion contributions ≤ 1.31 pp absolute, $\leq 13.2\%$ relative).

(iii) Dual filter dominates the faith loss in the ablation. Status: INVERTED. The empirical pattern is the

opposite — `sft_dual` is statistically indistinguishable from `sft_only` baseline; the faith loss is the load-bearing component (`sft_faith` alone gives +71.73 pp over baseline); the increment from adding the dual filter on top of the faith loss is +0.85 pp seed-mean (within one standard deviation of `sft_full`). We retain the dual filter in the recipe because the four-cell decomposition (Appendix B.2) demonstrates a symmetric-extension contribution axis that is not addressable by text- only filtering, and the full method’s +0.85 pp seed-mean over `sft_faith` is small but consistent across all three seeds.

Reporting commitment. One pre-registration HIT, one REFUTED, one INVERTED — all reported transparently. Each was tested independently in §??. The methodological refinement (`IMG_correct`) lands as a contribution independently of the empirical results, since it is a reusable audit-tool that improves the interpretation of any future or prior IMG-style metric.

G. Compute Disclosure

Total compute spent on this paper: $\approx$ \$400 on Modal H100 80 GB (approximately 30 H100-hours total — 15 Phase 4 fine-tunes at ~ 1.5 h each on simple-loss configurations and ~ 3 –4 h each on faith-loss configurations; 15 Phase 5 `IMG_correct` evals at 5–10 min each; 18 retroactive audit cells at 3–5 min each; 4 cross-counterfactual cells; 3 cross-site cells). All experiments are reproducible on a single H100 80 GB instance with the released code; the budget is well within reach for any researcher with comparable cloud- compute access. The full project envelope (including image-build retries, smoke tests, and the inevitable launcher overhead from two Modal billing-cycle cap-hit cycles during the run) is approximately \$250 over baseline; the headline experiments themselves are $\leq$ \$150.